8.2.3 Relation to Splitting Methods

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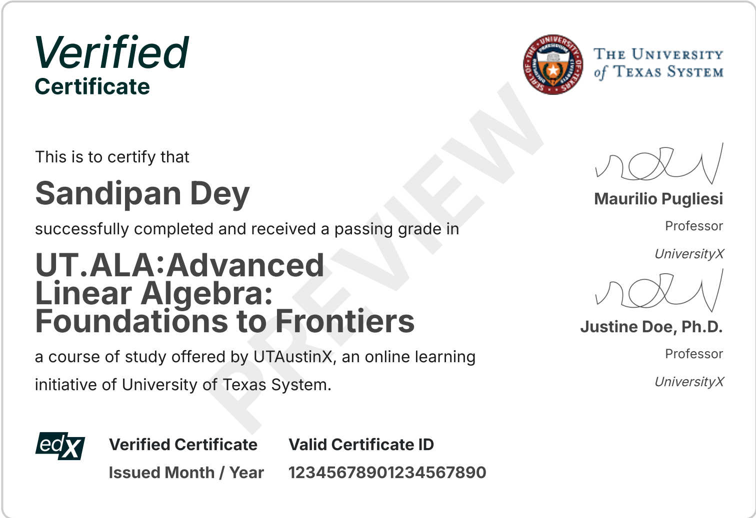

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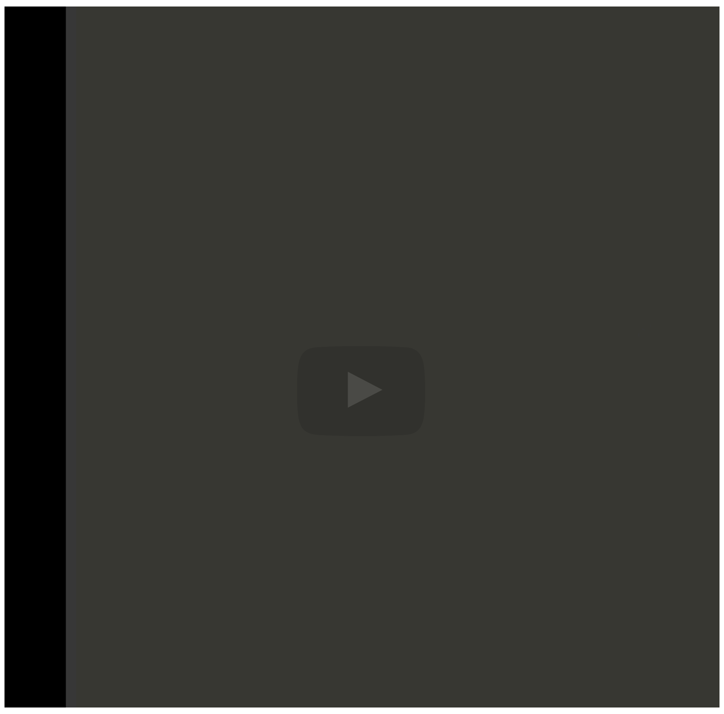

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2.0x

standard basis vectors. Remember these are the vectors where you

have a 1 that's the first standard basis vector. And then the second standard

basis vector and so forth. Now you're going to do a homework in which you show

that if you take the first standard basis vector to be your first search

direction, then you actually do exactly the updates that you did when you did

the Gauss-Seidel iteration. Hmm,

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Read Unit 8.2.3 of the notes. [LINK]

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? Cycling through standard basis vectors 2

Homework 8.2.3.1

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For

$$
\begin{array}{l}
\textbf{Given :}\ A, b, x^{(0)} \\
r^{(0)} := b - Ax^{(0)} \\
k := 0 \\
\textbf{while}\ r^{(k)} \neq 0 \\
\quad p^{(k)} := \text{ next direction} \\
\quad q^{(k)} := Ap^{(k)} \\
\quad \alpha_k := \frac{p^{(k)\,T} r^{(k)}}{p^{(k)\,T} q^{(k)}} \\
\quad x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)} \\
\quad r^{(k+1)} := r^{(k)} - \alpha_k q^{(k)} \\
k := k + 1 \\
\textbf{endwhile}
\end{array}
$$

show that if $p^{(0)} = e_0$, then

$$\chi_0^{(1)} = \chi_0^{(0)} + \frac{1}{\alpha_{0,0}} \left(\beta_0 - \sum_{j=0}^{n-1} \alpha_{0,j} \chi_j^{(0)} \right) = \frac{1}{\alpha_{0,0}} \left(\beta_0 - \sum_{j=1}^{n-1} \alpha_{0,j} \chi_j^{(0)} \right)$$

done

Solution:

For a complete solution see the notes.

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Answers are displayed within the problem

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So, what you notice is that you get the Gauss-Seidel iteration. It is not that

one iteration of this is equal to one iteration of the Gauss Seidel iteration.

No, instead n iterations, if a is n by n, n iterations of this is equivalent

to one iteration of Gauss-Seidel.

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